ChromIQ

Measurement Report - Verification of Profile - Demo-Report-Matrix-verify

Created: 2026-08-10 23:19:50

Report Scope

The following profile verification runs are included:

- Demo-Report-Matrix-verify, Instrument: X-Rite ColorMunki · 12 verification runs

No. of Measurements:
12

Date range:
2026-05-01 – 2026-08-10

Warning — mixed instruments. Every run in a report should come from the same instrument and the same printer; the report cannot tell printers apart. These runs use a different instrument from the majority (X-Rite ColorMunki):

- Demo-Report-Matrix-verify @ 2026-08-10T10:00:00 — uses X-Rite i1Pro3

Warning — missing cube colours. These runs are missing one or more of the eight cube corners, so their cube-corner figures are less meaningful:

- Demo-Report-Matrix-verify @ 2026-08-01T10:00:00 — missing White, Red, Green, Blue, Cyan, Magenta, Yellow
- Demo-Report-Matrix-verify @ 2026-08-08T10:00:00 — missing White, Red, Green, Blue, Cyan, Magenta, Yellow
- Demo-Report-Matrix-verify @ 2026-08-09T10:00:00 — missing White, Red, Green, Blue, Cyan, Magenta, Yellow

Warning — these verifications were not all printed the same way. A sheet printed through the profile measures the profile; a sheet printed raw measures the printer. The trend changes meaning at the point where the method changed:

- Demo-Report-Matrix-verify @ 2026-05-01T10:00:00 — printed raw — no profile
- Demo-Report-Matrix-verify @ 2026-06-01T10:00:00 — printed raw — no profile
- Demo-Report-Matrix-verify @ 2026-06-15T10:00:00 — printed through the profile
- Demo-Report-Matrix-verify @ 2026-07-01T10:00:00 — printed through the profile
- Demo-Report-Matrix-verify @ 2026-07-10T10:00:00 — printed in another app with colour management
- Demo-Report-Matrix-verify @ 2026-07-20T10:00:00 — method not recorded (made before ChromIQ recorded it, or printed outside ChromIQ)
- Demo-Report-Matrix-verify @ 2026-08-01T10:00:00 — printed raw — no profile
- Demo-Report-Matrix-verify @ 2026-08-08T10:00:00 — printed raw — no profile
- Demo-Report-Matrix-verify @ 2026-08-09T10:00:00 — method not recorded (made before ChromIQ recorded it, or printed outside ChromIQ)
- Demo-Report-Matrix-verify @ 2026-08-10T09:00:00 — printed in another app with colour management
- Demo-Report-Matrix-verify @ 2026-08-10T10:00:00 — printed raw — no profile
- Demo-Report-Matrix-verify @ 2026-08-10T11:00:00 — printed through the profile

How to read this report

This report compares what your instrument measured against the chart's design colours (the reference values the chart was built from). Every number is a colour difference (ΔE00): 0 is a perfect match, 1–2 is barely visible, and 10 or more is clearly different.

- Colour accuracy — the ΔE00 across the patches, split so you can see the bulk of the chart (all patches and the best 95%) apart from the few hardest patches (the worst 5%). Each metric is judged against your Pass thresholds.
- Paper white & darkest black — the brightest and deepest patches (L*), a quick health check of your paper and maximum ink.
- Cube corners — paper white, composite black and the six primary and secondary inks. These say as much about your inks as about the instrument.

What the numbers mean depends on how the chart was printed:

- A profiling chart is printed WITHOUT colour management (the raw print you measure to build a profile). It is not expected to match the design closely, so the ΔE can look large — that's normal. Here it is the CHANGE between dated reports that matters, not a single value.
- A verification chart is printed THROUGH your finished profile — ChromIQ converts the sheet itself and prints it with the printer's colour management off. It SHOULD match the design closely, so low ΔE and passes mean the profile is still accurate; rising numbers over time tell you when it's worth re-profiling. (Printed raw instead, the same sheet is a printer drift check — the report says which way each sheet was printed.)

Some of a chart's design colours can be brighter or more saturated than this printer and paper can physically produce — no profile can print them, however good it is. Where the report can tell (it asks the run's profile), it splits the colour-accuracy figures into two groups: “Within the profile's gamut” — the colours that were genuinely printable, the fair measure of accuracy — and “Beyond it” — the unreachable ones, whose distance describes the limit of the gamut, not a mistake of the profile. Every patch stays counted and visible; the Pass/Fail verdict judges the within-gamut figures.

Because the design reference never changes, comparing dated reports of the same chart on the same printer is a clean, reliable signal of drift — ageing inks, a wandering printer, or an instrument going off. Save a report after each measurement to build that history. Screen and print colours here are approximate; the numbers come from your measurement file and are exact.

Report Results

The following results are extracted from detailed data (Colour accuracy) for the included measurements in this report. Where a sheet's colours are split into within / beyond the profile's gamut, Pass and Fail judge the within-gamut figures — colours the profile could never print are not counted against it.

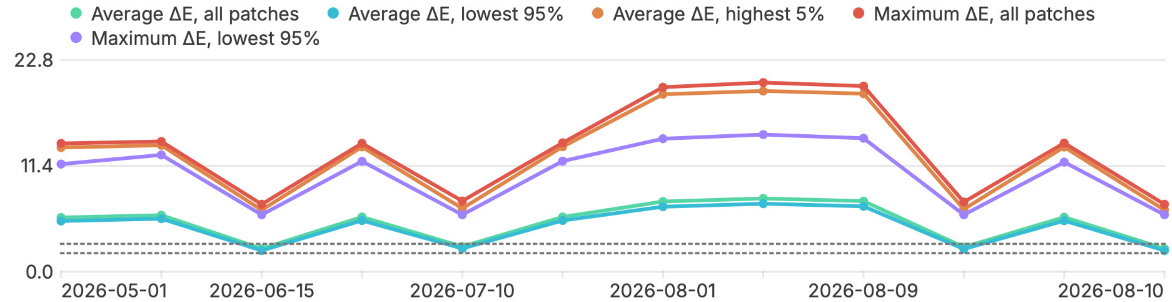
Metric	2026-05-01	2026-06-01	2026-06-15	2026-07-01	2026-07-10	2026-07-20
Average ΔE, all patches	Fail	Fail	Fail	Fail	Fail	Fail
Average ΔE, lowest 95%	Fail	Fail	Fail	Fail	Fail	Fail
Average ΔE, highest 5%	Fail	Fail	Fail	Fail	Fail	Fail
Maximum ΔE, all patches	Fail	Fail	Fail	Fail	Fail	Fail
Maximum ΔE, lowest 95%	Fail	Fail	Fail	Fail	Fail	Fail

Metric	2026-08-01	2026-08-08	2026-08-09	2026-08-10	2026-08-10	2026-08-10
Average ΔE, all patches	Fail	Fail	Fail	Fail	Fail	Fail
Average ΔE, lowest 95%	Fail	Fail	Fail	Fail	Fail	Fail
Average ΔE, highest 5%	Fail	Fail	Fail	Fail	Fail	Fail
Maximum ΔE, all patches	Fail	Fail	Fail	Fail	Fail	Fail
Maximum ΔE, lowest 95%	Fail	Fail	Fail	Fail	Fail	Fail

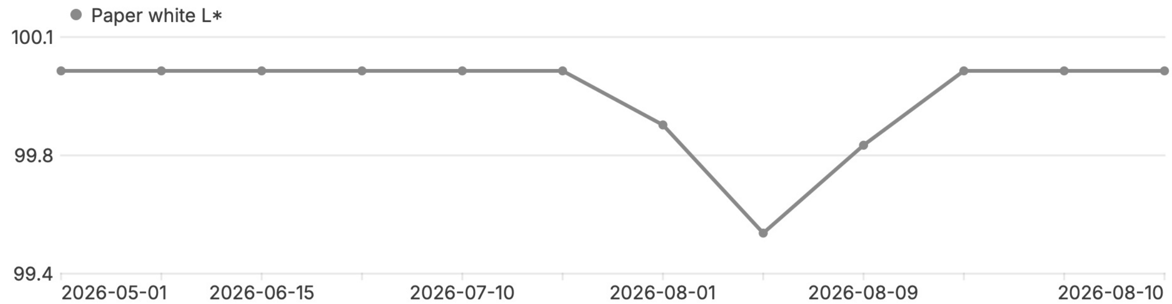
Trend over time (this printer)

A rising average or shifting white/black/colour over time points to ageing inks, printer drift, or instrument drift.

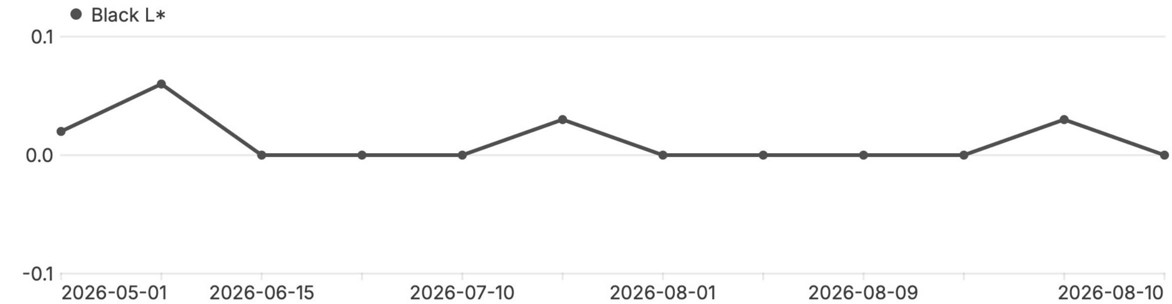
Colour accuracy (ΔE00)



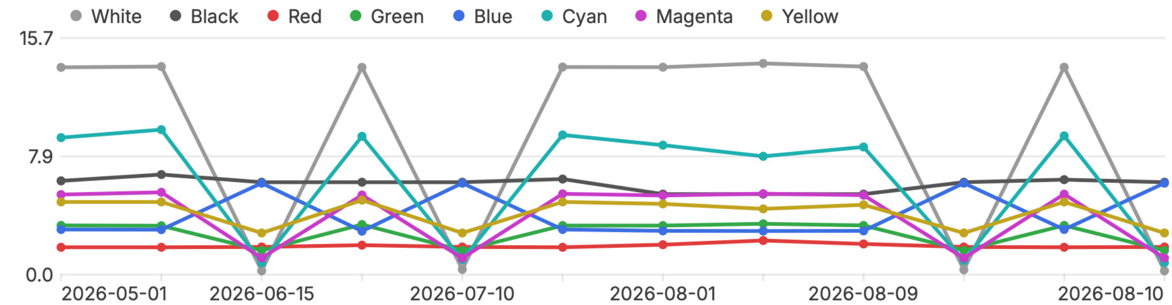
Paper white (L*)



Darkest black (L*)



Cube corners (ΔE00 per ink)



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Overview of Measurement Metrics

Metric	2026-05-01	2026-06-01	2026-06-15	2026-07-01	2026-07-10	2026-07-20
Within the profile's gamut						
Patches	97	97	97	97	97	97
Average ΔE, all patches	6.04	6.29	2.51	6.09	2.75	6.09
Average ΔE, lowest 95%	5.64	5.89	2.29	5.69	2.52	5.69
Average ΔE, highest 5%	13.38	13.62	6.65	13.45	6.82	13.46
Maximum ΔE, all patches	13.82	14.03	7.26	13.83	7.62	13.88
Maximum ΔE, lowest 95%	11.59	12.58	6.14	11.90	6.13	11.91
Beyond the profile's gamut						
Patches	8	8	8	8	8	8
Average ΔE, all patches	3.42	3.63	2.18	3.50	2.27	3.49
Average ΔE, lowest 95%	3.22	3.43	2.09	3.30	2.17	3.30
Average ΔE, highest 5%	4.83	5.04	2.77	4.95	2.95	4.83
Maximum ΔE, all patches	4.83	5.04	2.77	4.95	2.95	4.83
Maximum ΔE, lowest 95%	4.35	4.83	2.67	4.43	2.77	4.49
All patches together						
Average ΔE, all patches	5.84	6.09	2.49	5.89	2.71	5.90
Average ΔE, lowest 95%	5.46	5.71	2.28	5.52	2.50	5.52
Average ΔE, highest 5%	13.38	13.62	6.65	13.45	6.82	13.46
Maximum ΔE, all patches	13.82	14.03	7.26	13.83	7.62	13.88
Maximum ΔE, lowest 95%	11.59	12.58	6.14	11.90	6.13	11.91
Spread (std. dev.)	2.75	2.81	1.69	2.81	1.70	2.77
Paper white L*	100.0	100.0	100.0	100.0	100.0	100.0
Black L*	0.0	0.1	0.0	0.0	0.0	0.0
White ΔE00	13.77	13.82	0.26	13.77	0.35	13.79
Black ΔE00	6.23	6.65	6.14	6.14	6.14	6.35
Red ΔE00	1.82	1.82	1.84	1.96	1.84	1.82
Green ΔE00	3.27	3.25	1.61	3.33	1.61	3.26
Blue ΔE00	3.00	2.99	6.08	2.89	6.08	3.00
Cyan ΔE00	9.10	9.63	0.80	9.19	1.00	9.28
Magenta ΔE00	5.32	5.47	1.11	5.29	1.10	5.37
Yellow ΔE00	4.83	4.83	2.77	4.95	2.77	4.83

Metric	2026-08-01	2026-08-08	2026-08-09	2026-08-10	2026-08-10	2026-08-10
Within the profile's gamut						
Patches	106	106	106	97	97	97
Average ΔE, all patches	7.63	7.96	7.68	2.68	6.07	2.51
Average ΔE, lowest 95%	7.06	7.39	7.11	2.46	5.67	2.29
Average ΔE, highest 5%	19.11	19.46	19.17	6.74	13.43	6.65
Maximum ΔE, all patches	19.87	20.37	19.99	7.51	13.86	7.26
Maximum ΔE, lowest 95%	14.32	14.77	14.38	6.13	11.80	6.14
Beyond the profile's gamut						
Patches	2	2	2	8	8	8
Average ΔE, all patches	3.98	3.87	3.96	2.25	3.46	2.18
Average ΔE, lowest 95%	3.26	3.38	3.27	2.16	3.27	2.09
Average ΔE, highest 5%	4.70	4.37	4.64	2.85	4.83	2.77
Maximum ΔE, all patches	4.70	4.37	4.64	2.85	4.83	2.77
Maximum ΔE, lowest 95%	3.26	3.38	3.27	2.77	4.40	2.67
All patches together						
Average ΔE, all patches	7.56	7.89	7.61	2.65	5.87	2.49
Average ΔE, lowest 95%	7.00	7.33	7.05	2.44	5.50	2.28
Average ΔE, highest 5%	19.11	19.46	19.17	6.74	13.43	6.65
Maximum ΔE, all patches	19.87	20.37	19.99	7.51	13.86	7.26
Maximum ΔE, lowest 95%	14.32	14.77	14.38	6.13	11.80	6.14
Spread (std. dev.)	3.81	4.03	3.84	1.69	2.76	1.69
Paper white L*	99.8	99.5	99.8	100.0	100.0	100.0
Black L*	0.0	0.0	0.0	0.0	0.0	0.0
White ΔE00	13.78	14.03	13.82	0.33	13.78	0.26
Black ΔE00	5.35	5.35	5.35	6.14	6.31	6.14
Red ΔE00	1.99	2.27	2.04	1.84	1.82	1.84
Green ΔE00	3.26	3.38	3.27	1.61	3.27	1.61
Blue ΔE00	2.91	2.90	2.91	6.08	3.00	6.08
Cyan ΔE00	8.60	7.86	8.48	0.94	9.22	0.80
Magenta ΔE00	5.26	5.38	5.28	1.10	5.35	1.11
Yellow ΔE00	4.70	4.37	4.64	2.77	4.83	2.77